

Two pager on Panorama Localisation

The Challenge

Challenge name/tagline: Am I at a unique place? *Where in the world could here be?*

Challenge one-liner: Identify plausible locations of a given 360° panorama using interaction with AI agents.

Introduction

Panorama images offer a rich representation of the surroundings of any point on land since they, to a fair extent, remove the subjectivity of the picture framing by the photographer. Thus, a third person not involved in the image capture should be more capable of locating a 360° panorama than of a photo featuring a narrower field of view. The goal of the challenge is to find *one or multiple plausible locations* of daylight panoramas from real places, taken at no more than 5m above ground, without human image editing.

The Team will be handed two sets of panoramas to prepare the challenge. Some of them will be from easy to spot sites, others from less known places. The training images will not be localised and they will follow the same specifications of the scoring “run” images.

The Team may prepare the software or AI agents or whatever means it finds useful for the scoring “run” challenge on Sunday where a fresh set of panoramas will be handed. The second training set and the scoring run challenge are time limited.

The ideal process is that humans are so efficient at teaching the AI agent to solve the task that “it” can do the scoring “run” on its own. Otherwise, humans and the programmed AI agent will collaborate during the scoring “run” to get to *plausible locations*.

All interactions with the AI agents should be recorded in JSON format. Teams are encouraged to send their JSON logs to the Discord Channel to confirm that the format complies with the challenge rules. Albatroz will reply in shortest delay. During the training, multiple solutions can be submitted, especially in cases where a previous solution was found in breach of the rules.

The scoring run is divided in two stages:

- 1) Five panoramas + “Special One” panorama with a 3 x factor scoring.
- 2) One panorama solved in real time during the live presentation.

The winner of the Albatroz challenge will run against the winners of all other challenges solving a new, unknown, panorama in real time before the whole audience.

Multiple plausible locations are welcome, with a modest “cost” related to decreasing likelihood. However, if the AI agent trusts its findings, “it” can repeat one location along the list to improve its score.

The core of the challenge

The goal is to return a list of locations (in lat,lng coordinates) sorted by decreasing likelihood of a dated (YYYYMMDDhhmmss), equirectangular 360° panorama with 6080 x 3040 pixels in JPG format. The panoramas come mainly from Portugal but also from Spain and France.

Each Team will be handed two sets of 10 panoramas. Each panorama will be labelled as “Urban” or “non-Urban”. This label serves to select the appropriate distance function, θ_U or θ_{NU} .

The challenge includes a free training period on Saturday afternoon. At 19h00 (7pm) on Saturday, a second set of panoramas will be provided for an optional trial run, giving Teams 2 hours to test and evaluate their AI methods before the official scoring run on Sunday. Participation in this trial run is recommended, as it allows Teams to verify that their AI methods are working as intended and comply with the challenge rules.

Teams may continue improving their methods and repeating trials overnight. The true locations will not be disclosed, as the goal is to identify plausible locations. Team members should be able to assess whether a proposed location is plausible, even if its exact correctness cannot be confirmed.

The scoring run will take place on Sunday between 10h00 and 13h00, with each Team having a total of 60 minutes of solving time within this window.

The Team can use any tools during the scoring run on other equipment. However, it cannot feed data other than text and multiple choice “clicks” into the AI agent.

The scoring rules – which the Team is encouraged to apply to their AI agents - are as follows:

1. The *plausible location* list is limited to 10 entries; each entry contains only one (lat,lng) pair.
2. The geodesic distance between any candidate answer to the correct location is measured in meters using GIS.tools [1]. All distances below 10m will be considered exact: $d=0$.
3. The location at the top of the list is worth a base score of 1000 points that can be modified by multiplicative correction factors applied as stated next.
4. The distance to the goal¹ is a corrective factor θ_U or θ_{NU} , for urban and non-urban panoramas respectively, computed according to the formulas where d is the distance in meters:

$$\theta_U = \frac{2}{1 + \frac{d}{100}} \quad \theta_{NU} = \frac{2}{1 + \frac{d}{10000}}$$

5. Descending each rank in the result list will represent a 0.95 correction factor. Thus, the answer at the R^{th} rank starts at $1000 \times 0.95^{(R-1)}$.
6. The Team can write cues to the AI agents through prompts or other methods up to 1000 characters per cue. Direct or indirect references to locations or sites are excluded². For each cue the Team gives to the AI agent, it will represent a 0.8 correction factor.
7. The Team can answer multiple-choice questions proposed by the AI agent up to 4 options per question. These can be about features, about “geographic zones”, *etc.*. The only restriction is that they are generated by the AI agent and not “suggested” by the Team. Each multiple choice quiz asked by the AI agent will represent a 0.95 correction factor, regardless of the correctness of the answer.
8. In case the AI agent returns a text localization – e.g. “Praça do Comércio” or “Município de Beja” – the Team may define a geographic coordinate within the area with a correction factor of 0.5.
9. The correction factors for cues given according to #6, #7 and #8 are applicable to all subsequent candidate plausible locations for that panorama.
10. The “Special One” panorama is identified in the scoring “run”. It bears a **3 x** correction factor.
11. The score for each panorama will be the sum of the scores for each *plausible* location.
12. The final score of the Team will be the sum of the scores for each panorama. In case of tie, the Team which finishes in the shortest time wins.

Nota bene: differences and similarities with popular geo-referenced panorama games

While this challenge resembles GeoGuessr [2], WorldGuessr [3] or TimeGuessr [4] games or alike, some differences should be highlighted, especially for the benefit of regular players:

1. While these games feature panoramas from all over the world, in our case the geographic scope is narrower; hence the ambiguity or confusion between candidate locations should be higher.

¹ In many cases, the distance factor will remain unknown while all other correction factors are known.

² Example of unacceptable indirect reference: “the most famous sport here is big wave surfing’ referring to Nazaré.

2. While many panoramas are captured next to a street or road some of them are in open fields, so don't restrict your search to roadside image databases.
3. The mention of street or road names is rare in these panoramas, removing a powerful cue.

For the curious mind, the motivation for the challenge is AI related and explained in the companion document [5] that is not relevant to solve the challenge.

References

- [1] <https://gis.tools/distance-calculator>. This site uses the Haversine formula, see https://en.wikipedia.org/wiki/Haversine_formula
- [2] <https://www.geoguessr.com/>
- [3] <https://www.worldguessr.com/>
- [4] <https://timeguessr.com/>
- [5] Praying for Portugal – How the JunctionX challenge began

Future potential and benefits

This is a game, thus the immediate benefit is to have fun while playing the game.

The second level benefit is to foster interactive applications with AI agents where questions and answers come from both sides and humans learn and teach AI agents how each “side” observes one image looking for relevant elements.

The third and ultimate benefit is to identify which features are believed to be singular of landscapes and regions, and yet they resemble features found elsewhere; a more thorough research should highlight the minimal set of features that identify a single region beyond ambiguity.

After the challenge

The *ultimate purpose* of the challenge is to return the AI-assisted deduction process (could we call it *reasoning*?) that led to a set of possible locations ranked by likelihood and obtained from the cooperation of humans and AI agents.

One of the expected outcomes is to find pairs or multi-sets of places that are far apart but yet are quite similar in their landscape or views and probably share many environmental conditions, which are one of the root causes of landscape, the other being humans.

The challenge will be a full success if players conclude that regional affinities are stronger than differences and what fits one environment could also suit other environments and, finally, if one enjoys living *here*, one should also enjoy living *there*.